Qually User Manual

A Comprehensive Guide to Using the LLM Audit Tool for Social Science Research

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1 Introduction

This user manual provides step-by-step instructions for using Qually, an LLM Audit Tool designed for social science research. Qually enables researchers to conduct systematic evaluations of Large Language Models (LLMs) across multiple providers without requiring programming expertise.

1.1 What is Qually?

Qually is a graphical user interface (GUI)-based tool that allows researchers to:

- Conduct experiments across multiple LLM providers (OpenAI, Anthropic, Google, Mistral, Grok, DeepSeek)
- Compare model behaviors and responses
- Manage prompts and experimental conditions
- Export results for analysis

1.2 Who Should Use This Manual?

This manual is designed for:

- Social science researchers conducting LLM evaluations
- Researchers without programming experience
- Anyone needing systematic LLM auditing capabilities

2 Installation and Setup

2.1 Prerequisites

Before installing Qually, ensure you have:

- **Python 3.11 or higher**: Check your Python version by running:

```
python --version
```

- **pip package manager**: Usually included with Python
- **API keys**: Obtain API keys for the LLM providers you want to use:
  - OpenAI: `https://platform.openai.com/api-keys`
  - Anthropic: `https://console.anthropic.com/`
  - Google: `https://makersuite.google.com/app/apikey`
  - Mistral: `https://console.mistral.ai/`
  - Grok (xAI): `https://x.ai/`
  - DeepSeek: `https://platform.deepseek.com/`

2.2 Installation Steps

2.2.1 Step 1: Clone the Repository

Clone the Qually repository from GitHub:

```
git clone https://github.com/kartik0trivedi/Qually.git
cd Qually
```

2.2.2 Step 2: Install Dependencies

Install required Python packages:

```
pip install -r requirements.txt
```

The required packages include:

- PyQt6 (≥6.9.0) - GUI framework
- pandas (≥2.2.0) - Data manipulation
- requests (≥2.32.0) - HTTP requests
- cryptography (≥44.0.0) - API key encryption

2.2.3 Step 3: Run the Application

Launch Qually:

```
python qually_gui.py
```

2.3 First Launch

When you first launch Qually:

1. You'll see a welcome screen prompting you to select or create a project folder
2. Choose an existing folder or create a new one for your project
3. The application will initialize the project structure

3 Getting Started

3.1 Creating a Project

Qually uses a project-based organization system. Each project is stored in a folder containing:

- `api_keys.json` - Encrypted API keys (encrypted)
- `encryption_key.key` - Encryption key for API keys
- `prompts.json` - Your prompts
- `experiments.json` - Experiment definitions
- `results/` - Experimental results
- `data_files/` - Imported CSV files

3.1.1 Creating a New Project

1. On the welcome screen, click **"Select Project Folder"** or **"Create New Project"**
2. Choose a location for your project folder
3. Enter a project name
4. Click **"Create"** or **"Select"**

3.2 Adding API Keys

API keys are required to interact with LLM providers. Qually stores them securely using encryption.

3.2.1 Adding Your First API Key

1. Click on the **"API Keys"** tab
2. Select a provider from the list (e.g., OpenAI, Anthropic)
3. Enter your API key in the input field
4. Click **"Save"** or **"Test and Save"**
5. The key will be encrypted and stored locally

3.2.2 Security Notes

- API keys are encrypted using Fernet (symmetric encryption)
- Keys are stored locally in your project folder
- Never share your `api_keys.json` or `encryption_key.key` files
- Each project has its own encryption key

3.2.3 Supported Providers

Table 1 shows all supported providers and example models.

Provider	Example Models
OpenAI	gpt-4, gpt-3.5-turbo, gpt-4-turbo
Anthropic	claude-3-opus, claude-3.5-sonnet, claude-3-haiku
Google	gemini-1.5-pro, gemini-1.5-flash
Mistral	mistral-large-latest, open-mixtral-8x7b
Grok (xAI)	grok-1
DeepSeek	deepseek-chat, deepseek-coder

Table 1: Supported LLM Providers and Models

3.3 Importing Data

Qually supports importing prompts and data via CSV files.

3.3.1 CSV Format Requirements

Your CSV file should have the following structure:

- **Header row**: Contains column names
- **Data rows**: Each row represents a prompt or data entry
- **Required columns**: At minimum, a column containing your prompts
- **Optional columns**: Additional metadata (e.g., condition labels, categories)

3.3.2 Example CSV Format

```
prompt,condition,category
"Analyze␣the␣sentiment␣of␣this␣text:␣'I␣love␣this␣product!'",positive,
   sentiment
"Analyze␣the␣sentiment␣of␣this␣text:␣'This␣is␣terrible.'",negative,
   sentiment
```

3.3.3 Importing a CSV File

1. Click on the **"Data Files"** tab
2. Click **"Import CSV"** or **"Add Data File"**
3. Select your CSV file
4. Review the imported data in the table
5. Verify column mappings if prompted
6. Click **"Confirm Import"**

4 Working with Prompts

4.1 Creating Prompts

Prompts are the inputs you send to LLMs. You can create prompts individually or import them from CSV files.

4.1.1 Adding a Single Prompt

1. Navigate to the **"Prompts"** tab
2. Click **"Add Prompt"** or **"New Prompt"**
3. Enter your prompt text
4. Optionally add tags or categories
5. Click **"Save"**

4.1.2 Managing Prompts

The Prompts tab allows you to:

- View all prompts in a table
- Edit existing prompts
- Delete prompts
- Search/filter prompts
- Export prompts to CSV

4.2 Prompt Best Practices

- **Be specific**: Clear, specific prompts yield better results
- **Use examples**: Include examples when possible
- **Define format**: Specify the desired output format
- **Test incrementally**: Start with simple prompts, then refine

5 Running Experiments

5.1 Creating an Experiment

Experiments allow you to test multiple prompts across different LLM providers and models.

5.1.1 Basic Experiment Setup

1. Go to the **"Experiments"** tab
2. Click **"Create Experiment"** or **"New Experiment"**
3. Enter an experiment name
4. Add a description (optional but recommended)
5. Click **"Create"**

5.2 Adding Conditions

Conditions define different LLM configurations to test. Each condition specifies:

- Provider (e.g., OpenAI, Anthropic)
- Model (e.g., gpt-4, claude-3.5-sonnet)
- Parameters (temperature, max tokens, etc.)

5.2.1 Adding a Condition

1. In your experiment, click **"Add Condition"**
2. Enter a condition name (e.g., "GPT-4 Baseline")
3. Select a provider from the dropdown
4. Select a model from the dropdown
5. Configure parameters:
   - **Temperature**: Controls randomness (0.0-2.0)
   - **Max Tokens**: Maximum response length
   - **Top P**: Nucleus sampling parameter
   - **Frequency Penalty**: Reduce repetition
   - **Presence Penalty**: Encourage topic diversity
6. Click **"Add Condition"**

5.2.2 Example: Multi-Condition Experiment

A typical experiment might include:

- Condition 1: GPT-4 (temperature=0.7)
- Condition 2: Claude-3.5-Sonnet (temperature=0.7)
- Condition 3: Gemini-1.5-Pro (temperature=0.7)

This allows you to compare responses across providers.

5.3 Selecting Prompts

After creating an experiment, select which prompts to test:

1. In the experiment view, find the prompt selection section
2. Check the prompts you want to include
3. You can select all prompts or specific ones
4. Verify your selection

5.4 Executing an Experiment

Once your experiment is set up:

1. Review your experiment setup
2. Ensure all conditions have valid API keys
3. Click **"Run Experiment"** or **"Start"**
4. Monitor progress in the progress bar
5. View log messages for status updates

5.4.1 During Execution

While the experiment runs:

- A progress bar shows completion status
- Log messages indicate current activity
- The interface remains responsive
- You can cancel the experiment if needed

5.4.2 Rate Limiting

Qually automatically handles rate limiting:

- Default delays: 1-2 seconds between requests
- Provider-specific settings in `settings.json`
- Automatic retries on rate limit errors
- Progress tracking for long experiments

6 Results and Export

6.1 Viewing Results

After an experiment completes:

1. Navigate to the **"Results"** tab
2. Select your experiment from the list
3. View results in a table format
4. Columns include:
    - Prompt ID
    - Condition name
    - Provider and model
    - Response text
    - Timestamp
    - Parameters used

6.2 Filtering and Sorting

You can filter and sort results:

- **Filter by experiment**: Select specific experiments
- **Filter by condition**: View specific conditions
- **Filter by provider**: Focus on specific providers
- **Sort columns**: Click column headers to sort
- **Search**: Use search box to find specific prompts or responses

6.3 Exporting Data

Export your results for analysis in other tools (e.g., R, Python, Excel).

6.3.1 Export to CSV

1. In the Results tab, click **"Export"** or **"Download CSV"**
2. Choose export options:
   - Select experiments to include
   - Choose columns to export
   - Set filename
3. Click **"Export"**
4. Save the file to your desired location

6.3.2 CSV Export Format

The exported CSV includes:

- Experiment ID and name
- Condition details
- Prompt text
- Response text
- Timestamps
- Model parameters
- Metadata

6.3.3 Results File Structure

Results are stored in `results/` folder with filenames:

```
results/experiment_<ID>_<timestamp>.csv
```

7 Advanced Features

7.1 Project Management

7.1.1 Switching Projects

1. Go to **File** → **Open Project**
2. Select a different project folder
3. The application will load the new project

7.1.2 Backing Up Projects

To back up a project:

1. Copy the entire project folder
2. Include all subfolders (`results/`, `data_files/`)
3. Preserve `api_keys.json` and `encryption_key.key` for API key access

7.2 Customizing Rate Limits

You can adjust rate limiting in `settings.json`:

```
{
    "rate_limiting": {
        "openai": 1.0,
        "anthropic": 2.0,
        "google": 1.5,
        "mistral": 1.0,
        "grok": 1.0,
        "deepseek": 1.0
    }
}
```

Values are in seconds (minimum delay between requests).

7.3 Batch Processing

For large experiments:

- Qually handles batches automatically
- Progress is saved incrementally
- If interrupted, you can resume (results are saved)
- Monitor progress through the log panel

8 Troubleshooting

8.1 Common Issues

8.1.1 API Key Errors

Problem: "API key not found" or "Invalid API key"

Solutions:

- Verify API key is correctly entered
- Check that the API key is active
- Ensure you have credits/quota available
- Try testing the key in the provider's console

8.1.2 Rate Limit Errors

Problem: "Rate limit exceeded" or "429 errors"
Solutions:

- Increase rate limiting delays in `settings.json`
- Wait before resuming the experiment
- Check your API quota/limits
- Use fewer concurrent requests

8.1.3 Connection Errors

Problem: "Connection failed" or network errors
Solutions:

- Check your internet connection
- Verify firewall settings
- Try again after a few minutes
- Check provider status pages

8.1.4 Application Errors

Problem: Application crashes or freezes
Solutions:

- Check log files in `logs/` folder
- Restart the application
- Verify Python and package versions
- Report the issue with log files

8.2 Getting Help

- **GitHub Issues**: `https://github.com/kartik0trivedi/Qually/issues`
- **Documentation**: Check the main README.md
- **Email**: hello@kartiktrivedi.in
- **Website**: `https://www.kartiktrivedi.in`

9 Best Practices

9.1 Experimental Design

- **Plan your experiment**: Define research questions before starting
- **Start small**: Test with a few prompts before large experiments
- **Document conditions**: Use descriptive condition names
- **Keep notes**: Document any deviations or issues

9.2 Data Management

- **Organize prompts**: Use clear naming conventions
- **Back up regularly**: Copy project folders periodically
- **Version control**: Consider using Git for prompt management
- **Export results**: Export important results immediately

9.3 Reproducibility

- **Save experiments**: Keep experiment definitions
- **Document parameters**: Note all model parameters
- **Record timestamps**: Note when experiments were run
- **Version prompts**: Track prompt versions/changes

10 Appendix

10.1 Keyboard Shortcuts

- **Ctrl+N** (Cmd+N on Mac): New experiment
- **Ctrl+O** (Cmd+O on Mac): Open project
- **Ctrl+S** (Cmd+S on Mac): Save
- **Ctrl+E** (Cmd+E on Mac): Export results

10.2 File Structure Reference

```
project_folder/
  api_keys.json           # Encrypted API keys
  encryption_key.key      # Encryption key
  prompts.json            # Prompt definitions
  experiments.json       # Experiment definitions
  settings.json          # Application settings
  results/                # Experiment results
    experiment_*.csv
  data_files/            # Imported CSV files
    *.csv
  logs/                  # Application logs
    qually_gui.log
```

10.3 Model Parameter Guidelines

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Parameter	Typical Range	Description
Temperature	0.0-2.0	Higher = more random
Max Tokens	1-4096+	Maximum response length
Top P	0.0-1.0	Nucleus sampling
Frequency Penalty	-2.0-2.0	Reduce repetition
Presence Penalty	-2.0-2.0	Encourage diversity

Table 2: Model Parameter Guidelines

Version Information

This manual corresponds to Qually version 2.0.0 (LE2).